Data Visualization with Tableau

Question Bank: 2, 5 and 10 Mark Questions

Section	Marks each	Questions	Question numbers
Section 1	2	66	Q1 – Q66
Section 2	5	33	Q67 – Q99
Section 3	10	15	Q100 – Q114

Questions are grouped by marks and then by topic. Answers are in the separate **Answer Booklet**, which uses the same question numbers.

Section 1: 2-Mark Questions

Part A: Introduction to Data Visualization

Q1. Define data visualization.

Q2. List any four reasons why data visualization is important.

Q3. Name any four common types of charts and state what each is used for.

Q4. Who was William Playfair? What did he introduce?

Q5. What is a Coxcomb chart? Who used it and why?

Q6. How did John Snow use data visualization?

Q7. What was René Descartes' contribution to data visualization?

Q8. Name any four modern data visualization tools.

Q9. How do IT and cybersecurity teams use data visualization?

Q10. Mention any four professionals who use data visualization.

Part B: Getting Data into Tableau

Q11. What is a Tableau data source?

Q12. What is a data connector in Tableau?

Q13. Name the six types of data connectors in Tableau.

Q14. What are ODBC and JDBC?

Q15. List any four file types Tableau can connect to.

Q16. What is the Data Interpreter?

Q17. What is an RData file? Can Tableau connect to it directly?

Q18. What is Rserve?

Q19. What details are needed to connect Tableau to MySQL?

Q20. What is Custom SQL in Tableau?

Q21. What is a NoSQL database? Give two examples.

Q22. Give two differences between SQL and NoSQL databases.

Q23. Mention any four security measures used when Tableau connects to a database.

Part C: Live Connection, Extracts and In-Memory

Q24. What is a live connection?

Q25. What is an extract in Tableau?

Q26. What is the Hyper engine?

Q27. What is meant by in-memory processing in Tableau?

Q28. Differentiate full refresh and incremental refresh.

Q29. What is an extract filter? Give an example.

Q30. Give two advantages and two disadvantages of a live connection.

Part D: Organizing and Combining Data

Q31. What is the data layer in Tableau?

Q32. What is a table? Name its parts.

Q33. Differentiate logical tables and physical tables.

Q34. What is a union? Give an example.

Q35. Define dimensions with examples.

Q36. What is a discrete field? In which colour is it shown?

Q37. What are facts? Give examples.

Q38. What is a fact table?

Q39. What is a join? Name its types.

Q40. What is an inner join?

Q41. What is a relationship in Tableau?

Q42. What is data blending?

Q43. What are primary and secondary data sources in blending?

Q44. What is meant by level of detail?

Part E: Tableau Files

Q45. What is a Tableau workbook?

Q46. Differentiate .twb and .twbx.

Q47. What is a Tableau Packaged Data Source (.tdsx)?

Q48. Differentiate .tds and .tdsx.

Q49. What is a story in Tableau?

Part F: The Tableau Workspace

Q50. What is the Tableau workspace?

Q51. What is the Data Pane?

Q52. What is a calculated field? Give an example.

Q53. What is a hierarchy? Give an example.

Q54. What is a parameter?

Q55. Differentiate groups and sets.

Q56. What is the Analytics Pane?

Q57. Differentiate a constant line and a reference line.

Q58. What is a trend line? Name its types.

Q59. What is the use of the Swap button?

Q60. What is Show Me?

Q61. What is the use of Pause Updates and Run Update?

Q62. What is the Canvas in Tableau?

Q63. What is the Pages shelf?

Q64. What is a context filter?

Q65. What is the Marks Card?

Q66. Differentiate Keep Only and Exclude.

Section 2: 5-Mark Questions

Part A: Introduction to Data Visualization

Q67. Explain the importance and benefits of data visualization with an example.

Q68. Explain the common types of data visualization charts.

Q69. Write a short note on the contributions of Florence Nightingale, John Snow and Charles Minard.

Q70. Explain who uses data visualization and how (any five users).

Part B: Getting Data into Tableau

Q71. Explain the steps to get data into Tableau.

Q72. Explain the types of data sources Tableau can connect to.

Q73. Explain the types of data connectors in Tableau.

Q74. Explain the steps to connect a text (CSV) file to Tableau. Mention its advantages and limitations.

Q75. How can data stored in an RData file be used in Tableau? Explain.

Q76. Explain the steps to connect Tableau with a MySQL database.

Q77. Differentiate a database and a Tableau data source.

Q78. Explain NoSQL databases and how Tableau connects to them.

Part C: Live Connection, Extracts and In-Memory

Q79. Explain live connection with its advantages and disadvantages.

Q80. Differentiate live connection and extract.

Q81. Explain in-memory processing in Tableau with its advantages and limitations.

Q82. Explain data extraction in Tableau and the steps to create an extract.

Part D: Organizing and Combining Data

Q83. Explain the components of the data layer in Tableau.

Q84. Explain the types of tables in Tableau and the ways to combine them.

Q85. Explain dimensions and their purpose in Tableau with an example.

Q86. Explain facts with a fact table example. How are facts related to measures?

Q87. Differentiate facts and dimensions.

Q88. Differentiate joins and relationships in Tableau.

Q89. Differentiate data blending and joins.

Part E: Tableau Files

Q90. Explain the components of a Tableau workbook and its file types.

Q91. Explain the Tableau Packaged Data Source (.tdsx) with its components, advantages and limitations.

Part F: The Tableau Workspace

Q92. Explain any five menus of Tableau Desktop.

Q93. Explain the important toolbar buttons in Tableau.

Q94. Differentiate the Data Pane and the Analytics Pane.

Q95. Explain the features of the Canvas in Tableau.

Q96. Differentiate the Columns shelf and the Rows shelf with an example.

Q97. Explain the Filters shelf and the types of filters in Tableau.

Q98. Differentiate the Filters shelf and the Pages shelf.

Q99. Explain the features of the Analytics Pane.

Section 3: 10-Mark Questions

Part A: Introduction to Data Visualization

Q100. What is data visualization? Explain why data is visualized, its benefits, common types and applications.

Q101. Explain the history of data visualization from ancient times to the modern era.

Part B: Getting Data into Tableau

Q102. Explain the complete process of getting data into Tableau, including types of data connections and connection modes.

Q103. Explain data connectors in Tableau: their types, how they work and the connection modes.

Q104. Explain databases in Tableau: types, connecting, live vs extract, working with tables, custom SQL and security.

Part C: Live Connection, Extracts and In-Memory

Q105. Compare live connection, extract and in-memory processing in Tableau in detail.

Part D: Organizing and Combining Data

Q106. What is a join? Explain the types of joins in Tableau with examples and a diagram.

Q107. Explain data blending in detail. How is it different from joins?

Q108. Explain facts and dimensions with a fact table and dimension table (star schema) example.

Q109. Explain the data layer and tables in Tableau, including relationships, joins and unions.

Part E: Tableau Files

Q110. Explain Tableau workbooks and packaged files (.twb, .twbx, .tds, .tdsx, .hyper).

Part F: The Tableau Workspace

Q111. Explain the Tableau workspace and its components with a neat diagram.

Q112. Explain the Data Pane and its features in detail.

Q113. Explain the Analytics Pane: its features, how to use it and the types of analysis it supports.

Q114. Explain the Columns, Rows, Filters and Pages shelves in Tableau with examples.